Instructor's Solution Manual: Chapter 2 - Section F

Applications on the Conservation of Linear Momentum

Context: These solutions cover Section F of the "Worksheet Chapter 2 G12" which was missing from the uploaded Answer Key.

Application 1: Elastic Collision

Problem Analysis: Two particles S_1 and S_2 collide elastically.

- **System:** $\{S_1, S_2\}$
- **Condition:** Elastic Collision $\implies$ Conservation of Linear Momentum ($\vec{P}$) AND Conservation of Kinetic Energy (KE).

Given:

- $m_1 = 100 \text{ g} = 0.1 \text{ kg}$
- $m_2 = 50 \text{ g} = 0.05 \text{ kg}$
- $V_2 = 1 \text{ m/s}$ (Assuming same direction or opposite depends on vector definition, standard derivation assumes vectors).

a) Determine the velocities $\vec{V_1'}$ and $\vec{V_2'}$ just after collision.

From the "Chapter 2: Linear Momentum" notes (Section F.1), for a 1D elastic collision, the final velocities are derived as:

$$V_1' = \frac{(m_1 - m_2)V_1 + 2m_2V_2}{m_1 + m_2}$$

$$V_2' = \frac{(m_2 - m_1)V_2 + 2m_1V_1}{m_1 + m_2}$$

Note: Students should simply apply these derived formulas or set up the conservation system:

1. $m_1V_1 + m_2V_2 = m_1V_1' + m_2V_2'$
2. $\frac{1}{2}m_1V_1^2 + \frac{1}{2}m_2V_2^2 = \frac{1}{2}m_1V_1'^2 + \frac{1}{2}m_2V_2'^2$

b) Determine the force $\vec{F}_{1/2}$ exerted by S_1 on S_2.

- **Concept:** Impulse-Momentum Theorem applied to particle S_2.
- **Time interval:** $\Delta t = 0.05 \text{ s}$

$$\Delta\vec{P_2} = \vec{F}_{1/2} \cdot \Delta t$$

$$\vec{P_2'} - \vec{P_2} = \vec{F}_{1/2} \cdot \Delta t$$

$$m_2(\vec{V_2'} - \vec{V_2}) = \vec{F}_{1/2} \cdot \Delta t$$

$$\vec{F}_{1/2} = \frac{m_2(V_2' - V_2)}{\Delta t}\hat{i}$$

c) Deduce $\vec{F}_{2/1}$.

- **Concept:** Newton's Third Law (Action-Reaction).

$$\vec{F}_{2/1} = -\vec{F}_{1/2}$$

Application 2: Jumping off a boat (Explosion)

Problem Analysis: A boy jumps off a boat. Initially, the system is at rest.

- **System:** $\{Boy, Boat\}$
- **Initial State:** $\vec{V}_{syst} = \vec{0} \implies \vec{P_i} = \vec{0}$

a) Determine the boat's velocity $\vec{V}$ in terms of m (boy), M (boat), and $\vec{v}$ (boy's jump velocity).

Apply Conservation of Linear Momentum (Isolated System, $\sum \vec{F}_{ext} = \vec{0}$):

$$\vec{P_i} = \vec{P_f}$$

$$\vec{0} = \vec{P}_{boy} + \vec{P}_{boat}$$

$$\vec{0} = m\vec{v} + M\vec{V}$$

Solving for $\vec{V}$:

$$M\vec{V} = -m\vec{v}$$

$$\vec{V} = -\frac{m}{M}\vec{v}$$

Interpretation: The boat moves in the opposite direction to the boy's jump.

b) Is the explosion elastic?

- **Definition:** An interaction is elastic if Kinetic Energy is conserved ($KE_i = KE_f$).
- **Check:**
  - $KE_i = 0$ (System at rest).
  - $KE_f = \frac{1}{2}mv^2 + \frac{1}{2}MV^2 > 0$.
- **Conclusion:** Since $KE_f \neq KE_i$, the explosion is **not elastic**. It is a "super-elastic" event where potential chemical energy (muscle) is converted into kinetic energy.

Application 3: Explosion of two pucks

Problem Analysis: Two pucks connected by a compressed spring are released.

- **Given:**
  - $m_A = 100 \text{ g} = 0.1 \text{ kg}$
  - $m_B = 50 \text{ g} = 0.05 \text{ kg}$
  - Spring compression: $\Delta x = L_0 - L = 10 - 5 = 5 \text{ cm} = 0.05 \text{ m}$
  - Velocity of A after explosion: $V_1 = 1 \text{ m/s}$ (Assuming to the left/negative based on typical diagrams, or strictly magnitude).

a) Determine $\vec{V_2}$ (Velocity of Puck B).

Apply Conservation of Linear Momentum ($\vec{P_i} = \vec{P_f} = \vec{0}$):

$$\vec{0} = m_A\vec{V_1} + m_B\vec{V_2}$$

$$m_B\vec{V_2} = -m_A\vec{V_1}$$

$$\vec{V_2} = -\frac{m_A}{m_B}\vec{V_1}$$

$$\vec{V_2} = -\frac{0.1}{0.05}(1\hat{i}) = -2(1\hat{i})$$

If $\vec{V_1}$ is direction $\hat{i}$, then $\vec{V_2} = -2\hat{i} \text{ m/s}$. (Speed $V_2 = 2 \text{ m/s}$ in opposite direction).

b) Is the explosion elastic?

- $KE_i = 0$
- $KE_f = \frac{1}{2}m_AV_1^2 + \frac{1}{2}m_BV_2^2 > 0$
- **Conclusion:** No. KE increases. The energy comes from the potential elastic energy of the spring.

c) Determine the spring constant K.

Apply Conservation of Energy (assuming no friction during expansion):

$$E_{mech(initial)} = E_{mech(final)}$$

$$HPE_{spring} = KE_{total}$$

$$\frac{1}{2}k(\Delta x)^2 = \frac{1}{2}m_AV_1^2 + \frac{1}{2}m_BV_2^2$$

Substitute values:

$$k(0.05)^2 = (0.1)(1)^2 + (0.05)(2)^2$$

$$0.0025k = 0.1 + 0.2$$

$$0.0025k = 0.3$$

$$k = \frac{0.3}{0.0025} = 120 \text{ N/m}$$

Application 4: Colliding objects stick together (Perfectly Inelastic)

Problem Analysis:

- **Given:**
  - $m_1 = 700 \text{ g} = 0.7 \text{ kg}$, $V_1 = 1 \text{ m/s}$
  - $m_2 = 80 \text{ g} = 0.08 \text{ kg}$, $V_2 = 5 \text{ m/s}$
  - *Note: Directions are crucial. Usually, collision implies opposite directions. Let's assume $\vec{V_1} = 1\hat{i}$ and $\vec{V_2} = -5\hat{i}$.*

a) Determine $\vec{V}_{syst}$.

Apply Conservation of Momentum:

$$m_1\vec{V_1} + m_2\vec{V_2} = (m_1 + m_2)\vec{V}_{syst}$$

$$\vec{V}_{syst} = \frac{m_1V_1 + m_2V_2}{m_1 + m_2}$$

Scenario A (Head-on):

$$\vec{V}_{syst} = \frac{(0.7)(1) + (0.08)(-5)}{0.7 + 0.08}$$

$$\vec{V}_{syst} = \frac{0.7 - 0.4}{0.78} = \frac{0.3}{0.78} \approx +0.38 \text{ m/s}$$

(Direction of mass 1).

b) Does the KE increase or decrease?

- **General Rule:** In inelastic collisions (sticking together), max kinetic energy is lost to deformation/heat.
- **Calculation:**
  - $KE_i = \frac{1}{2}(0.7)(1)^2 + \frac{1}{2}(0.08)(5)^2 = 0.35 + 1.0 = 1.35 \text{ J}$
  - $KE_f = \frac{1}{2}(0.78)(0.38)^2 \approx 0.056 \text{ J}$
- **Conclusion:** KE **decreases** drastically. It is converted into internal energy (deformation/heat).